

Calculating Volatility in Frontier Markets Without Options

Evidence and a Practical Framework from the Nepal Stock Exchange

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Abstract

Options markets make volatility visible in a way that cash-only markets do not: option prices can be inverted to obtain a forward-looking, risk-neutral measure of expected volatility. Nepal has no exchange-traded equity options or futures, so there is no NEPSE analogue of the VIX that can be read directly from an option chain. This paper asks what can be measured instead, and how reliable those measurements are in a frontier-market setting. Using daily open, high, low, and close (OHLC) data from the Nepal Stock Exchange, the study evaluates close-to-close volatility and the Parkinson, Garman-Klass, Rogers-Satchell, Yang-Zhang, and additive Rogers-Satchell approaches. The main empirical sample contains 143,149 stock-days for 291 ordinary equities from March 2024 through August 2026. The results show that daily ranges contain useful information even in Nepal: among ordinary equities, the Parkinson standard-deviation ratio relative to a matched open-to-close benchmark remains between 0.935 and 1.058 across liquidity groups, and the thinnest equity group has a zero-range rate of only 1.63%. A cross-market check provides an external validation anchor rather than an implied-volatility substitute. On the NIFTY 50, using the same 21-session variance aggregation specified in this paper, Parkinson volatility has a correlation of 0.776 with India VIX and an R^2 of 0.602; standard 21-session close-to-close volatility has a correlation of 0.796 and an R^2 of 0.634. Both therefore co-move strongly with the options-derived volatility state, but Parkinson does not outperform close-to-close in this validation. The paper also shows why frontier-market implementation requires more than choosing a formula. Pooling debentures, funds, promoter shares, and ordinary equity creates a false appearance of estimator breakdown; the thinnest pooled group is dominated by non-equity securities and produces a Rogers-Satchell variance ratio of only 0.172. The practical conclusion is that a cash-only frontier market can estimate historical and realized volatility credibly from daily OHLC data, but it cannot recover option-implied volatility from spot prices alone. Reliable measurement therefore depends on instrument classification, session-aware data cleaning, liquidity diagnostics, appropriate treatment of overnight returns, and validation of any bias correction before it is applied.

Keywords: frontier markets; volatility; Nepal Stock Exchange; range-based estimators; implied volatility; Parkinson; Rogers-Satchell; market microstructure

1. Introduction

Volatility is one of the basic quantities in finance. It enters portfolio allocation, risk limits, Value at Risk, position sizing, forecasting, and derivative pricing. In developed markets, investors can observe both realized price variation and a forward-looking volatility signal embedded in option prices. The best-known example is the VIX, which uses S&P 500 option prices to summarize the market's risk-

neutral expectation of near-term variance. Frontier markets often do not have this second source of information. As of 2026, the Nepal Stock Exchange (NEPSE) remains a cash market without exchange-traded stock or index options and futures, so there is no option chain from which to infer an implied-volatility surface or a Nepalese VIX-style index (The Kathmandu Post, 2026).

The absence of options creates a measurement problem, but it does not make volatility unmeasurable. It changes the object that can be estimated. From cash-market prices, researchers can estimate historical or realized volatility: the amount of variation that actually occurred in observed prices. They cannot, from those prices alone, recover the same risk-neutral, forward-looking expectation that an option-implied measure contains. This distinction is central. A good historical estimator can be an informative risk measure and a forecasting input, but it is not a synthetic option market.

The simplest historical estimator uses the standard deviation of close-to-close returns. That approach is easy to calculate, but it discards most of the information contained in each trading session. Range-based estimators use the daily high and low, and in some cases the open and close, to extract more information from the same daily bar. Parkinson (1980), Garman and Klass (1980), Rogers and Satchell (1991), and Yang and Zhang (2000) provide a family of estimators built for this purpose. These methods are attractive for a market such as Nepal because daily OHLC data are much easier to obtain consistently than tick-level or high-frequency data.

However, frontier markets introduce complications that are less important in dense developed markets. A reported daily high and low may be based on only a handful of transactions. Stale prices can create zero returns or zero ranges. Price limits and auction rules can constrain observed prices or shift where variation appears in the daily bar. The trading week can change. Most importantly in the NEPSE data used here, the exchange's daily file places ordinary equities, corporate debentures, closed-end funds, and restricted promoter shares in the same table without an instrument-type field. A mechanically constructed 'thin-market' sample can therefore become a sample of different asset classes rather than a sample of thinly traded common stocks.

This paper makes two contributions. First, it develops a practical framework for estimating volatility in a frontier market that lacks options, with explicit formulas, benchmark choices, annualization rules, and diagnostic checks. Second, it applies that framework to NEPSE data and shows which concerns matter empirically. The results are encouraging for ordinary equity: range estimators do not collapse even in the least active equity group. At the same time, the results are cautionary about sample construction and additive bias corrections. The correct question is not simply 'Which volatility formula should Nepal use?' but 'Which volatility object is being measured, from which securities, over which part of the day, and under what market microstructure?'

2. What Changes When There Is No Options Market?

In a market with actively traded options, implied volatility is obtained by taking an option pricing relation and solving for the volatility parameter that makes the model price equal the observed market price. The result is forward-looking in the sense that it reflects prices for future-contingent claims. It is also a risk-neutral quantity: it embeds both expected future variation and the compensation investors

demand for bearing volatility risk. Historical volatility, by contrast, is computed from realized spot-market prices and is a statistical property of past returns.

The two measures are related but not interchangeable. A historical estimator can track the same volatility state that influences option prices, yet it need not equal implied volatility because option prices contain expectations and risk premia. This is why the cross-market India VIX exercise in Section 6 is treated as a validation anchor rather than an identity test. A strong correlation supports the usefulness of OHLC-based volatility as a state variable; it does not turn it into implied volatility.

For a cash-only market, the practical objective should therefore be stated as follows: estimate the current and recent variance of the traded price process as efficiently and robustly as possible, then use a separate forecasting model if a forward-looking estimate is needed. The measurement layer and the forecasting layer should not be conflated. The present paper concentrates on the measurement layer.

3. Data and NEPSE Market Setting

The main analysis panel, after the paper's positivity, return, and feasible-range screens, covers March 4, 2024 through August 26, 2026 and contains 569 trading sessions and 520 securities. The immediately preceding cleaned trading panel contains 521 securities; one security drops out under the analysis screens. After instrument classification, 291 ordinary equities contribute 143,149 stock-days. The pooled analysis file also contains closed-end mutual funds, corporate debentures, and restricted promoter shares. A separate NEPSE index series provides daily OHLC observations, while NIFTY 50 OHLC data and India VIX provide an external developed-market comparison over 2010-2026.

The cleaning protocol is deliberately conservative. An OHLC record must satisfy $\text{High} \geq \max(\text{Open}, \text{Close})$ and $\text{Low} \leq \min(\text{Open}, \text{Close})$. When an envelope inconsistency occurs, the high and low are widened to include the reported open and close while the original values are retained and the record is flagged. Duplicate security-date observations that conflict on OHLC values are excluded rather than resolved by file order. In the final ordinary-equity sample, only 122 rows, approximately 0.09%, require OHLC envelope repair.

Duplicate reconciliation is explicit as well. The historical long-form source panel contains 640 duplicated (security, date) keys: 622 exact-duplicate keys comprising 1,296 physical rows and 18 keys with conflicting OHLC values comprising 36 rows. Exact duplicates are collapsed to one record, removing 674 redundant rows; conflicting-OHLC keys are excluded in full, removing 36 rows rather than choosing a record by file order. The reconciliation script verifies that the exact duplicates also agree on volume and turnover. In total, the duplicate rule removes 710 physical rows from that raw long-form panel. This duplicate audit is separate from the 2024-2026 daily-trades panel used for the paper's main equity estimates.

The calendar also matters. NEPSE historically traded Sunday through Thursday and moved to a Monday-through-Friday schedule during the April 2026 rule transition. The repository records April 20, 2026 as the documented regime date, while the data-driven session detector retains off-schedule transition sessions rather than forcing a hard calendar break. The project therefore detects genuine

sessions from the data and cross-checks them against the documented weekday regime rather than hard-coding a fixed calendar. This matters for overnight returns and annualization. A fixed 252-day convention, standard in many developed-market applications, should not be imposed automatically on a market with a different holiday and session structure.

Table 1. Data used in the empirical analysis

Dataset	Period / size	Use in this paper
NEPSE main analysis panel (after screens)	2024-03-04 to 2026-08-26; 520 securities	Main OHLC and liquidity analysis
NEPSE ordinary-equity subset	291 equities; 143,149 stock-days	Primary estimation universe
NEPSE index	2,296 usable sessions	Aggregate-market comparison
NIFTY 50	4,037 sessions, 2010-2026	Cross-market implementation check
India VIX	4,036 raw; 3,999 after 21-session matching	Options-based external validation anchor

4. Volatility Estimators Available Without Options

Let O_t , H_t , L_t , and C_t denote the opening, high, low, and closing price for trading session t , and let C_{t-1} be the previous genuine session's close. All formulas below first estimate daily variance. A rolling volatility measure is then formed by averaging daily variance over a chosen window and taking the square root. Annualization should use A , the number of genuine exchange sessions per year or an explicitly defined calendar estimate, rather than assuming $A = 252$.

4.1 Close-to-close volatility

The standard baseline uses log close-to-close returns $r_t = \ln(C_t/C_{t-1})$. For a window of w sessions, historical volatility is the sample standard deviation of those returns.

$$\hat{\sigma}_{CC,annual} = \sqrt{A} \times SD(r_t \text{ over the previous } w \text{ sessions})$$

Its advantage is robustness to errors in the reported high and low. Its disadvantage is statistical inefficiency: it uses one endpoint movement and ignores the intraday path.

4.2 Parkinson estimator

Parkinson (1980) uses only the daily high-low range. Under its maintained continuous-path, zero-drift model, the range contains substantially more information about variance than a single close-to-close return.

$$\hat{v}_{P,t} = [\ln(H_t/L_t)]^2 / (4 \ln 2)$$

For a rolling window, annualized volatility can be reported as $\sqrt{A \times \text{mean}(\hat{v}_{P,t})}$. In a thin market, the main danger is that the observed high and low are the extremes of a small number of trades rather than the extremes of a continuously observed path. If $H_t = L_t$, the Parkinson estimate is exactly zero.

4.3 Garman-Klass estimator

The simplified Garman-Klass estimator combines the range with the open-to-close move:

$$\hat{v}_{GK,t} = 0.5[\ln(H_t/L_t)]^2 - (2 \ln 2 - 1)[\ln(C_t/O_t)]^2$$

It can be more efficient than close-to-close variance when its assumptions are reasonable. Importantly, on a valid OHLC bar this simplified expression is non-negative. A negative estimate is therefore a data-quality warning rather than an economic feature of thin trading.

4.4 Rogers-Satchell estimator

Rogers and Satchell (1991) construct an estimator that is independent of drift under a continuously observed Brownian-motion model. Define $u_t = \ln(H_t/O_t)$, $d_t = \ln(L_t/O_t)$, and $c_t = \ln(C_t/O_t)$. Then

$$\hat{v}_{RS,t} = u_t(u_t - c_t) + d_t(d_t - c_t)$$

The drift-independence is useful in equity markets, but it does not make the estimator immune to discretization. If very few trades occur, the reported extrema can still miss path variation. Rogers-Satchell can also equal zero on monotone sessions even when the price moved, which is a known property of the formula rather than automatically a bad record.

4.5 Yang-Zhang and overnight variation

The Parkinson, Garman-Klass, and Rogers-Satchell daily formulas primarily describe the within-session path. If the research objective is total daily risk, overnight gaps must also be included. Yang and Zhang (2000) combine an overnight variance term, an open-to-close variance term, and a Rogers-Satchell component. With an n -session window, its weight is $k = 0.34/[1.34 + (n+1)/(n-1)]$. The estimator is attractive when overnight price discovery is economically important, but it requires a correct session calendar because the 'previous close' must mean the previous genuine trading session.

4.6 Bias-corrected range estimators

Maheswaran and Kumar (2013) and Kumar and Maheswaran (2014) develop corrections intended for downward bias caused by discrete observations of the price path. The additive Rogers-Satchell implementation studied in the NEPSE project can be written as Rogers-Satchell plus a non-negative correction tied to boundary cases in which the high or low coincides with the open or close. Such a correction is valuable only if the uncorrected estimator is actually downward biased for the target market and benchmark. The empirical results below show why this premise must be tested rather than imported from another market.

Table 2. Practical interpretation of cash-market volatility estimators

Estimator	Uses	Main strength	Key frontier-market risk
Close-to-close	C_t, C_{t-1}	Simple; robust to H/L errors	Discards intraday information; stale closes
Parkinson	H_t, L_t	Efficient use of range	Zero/understated range with few trades
Garman-Klass	O, H, L, C	Uses range plus O-C move	Opening jumps; invalid OHLC

			if not cleaned
Rogers-Satchell	O,H,L,C	Drift-independent under maintained model	Discrete extrema; monotone-day zeros
Yang-Zhang	Prior C + OHLC	Includes overnight and intraday variation	Calendar definition; rolling-window dependence
AddRS	OHLC + boundary correction	Targets discrete-extrema bias	Can overcorrect when base estimator is not downward biased

5. Frontier-Market Problems That Must Be Solved Before Calculation

5.1 Instrument composition

The largest empirical warning in the NEPSE project is not a volatility formula but a sample-definition problem. The published daily file contains several security classes without an instrument-type field. When all 520 securities are pooled and sorted by trading intensity, the thinnest quintile has median trading intensity of only two trades per day, participation of 19.5%, $P(H=L)$ of 54.6%, and a Rogers-Satchell-to-benchmark variance ratio of 0.172. That looks like a dramatic failure of range-based estimation.

But the liquidity ranking is largely an asset-class ranking. The two thinnest pooled quintiles contain only four ordinary equities out of 209 securities. The thinnest pooled decile is 94.3% non-equity by stock-day. Corporate debentures and restricted promoter shares are structurally different instruments; treating them as thin common stocks creates a composition artifact.

A frontier liquidity gradient that is really an asset-class gradient

NEPSE publishes one daily file for every listed instrument and no instrument-type field. Sorting it by trading intensity yields a strong, clean, spurious gradient: the thin end is debentures, closed-end funds and restricted promoter shares. Within ordinary equity the gradient is absent.

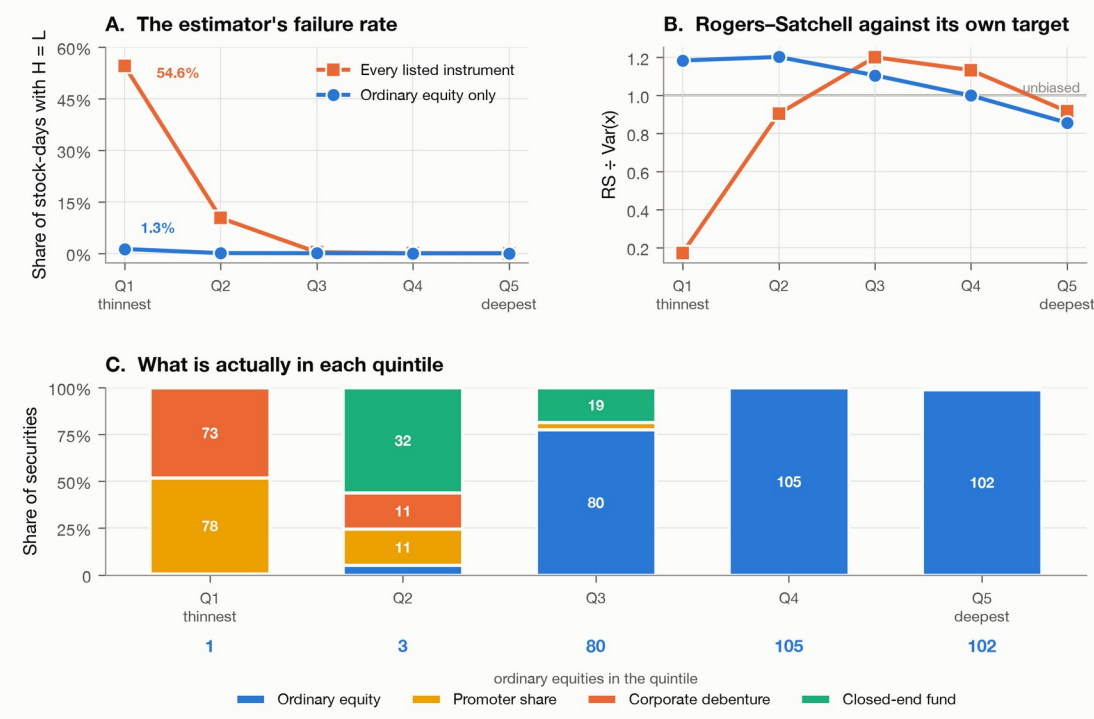


Figure 1. Instrument composition of pooled liquidity quintiles. The apparently most illiquid buckets are overwhelmingly non-equity.

5.2 Sparse trading and stale observations

Sparse trading affects both endpoints and extremes. A stale close can produce a zero close-to-close return even when the latent value changed, while a small number of transactions can produce a high and low that cover only part of the true path. These mechanisms can bias volatility downward. At the same time, bid-ask bounce and price impact can widen observed extremes and bias range measures upward. Thin-market bias therefore need not be one-directional; it should be diagnosed against liquidity and with cross-market or simulation checks.

Trading intensity in the pooled universe, before any instrument filter

Every listed instrument type · 184,390 stock-days · 520 securities · 2024-03-04 to 2026-08-26

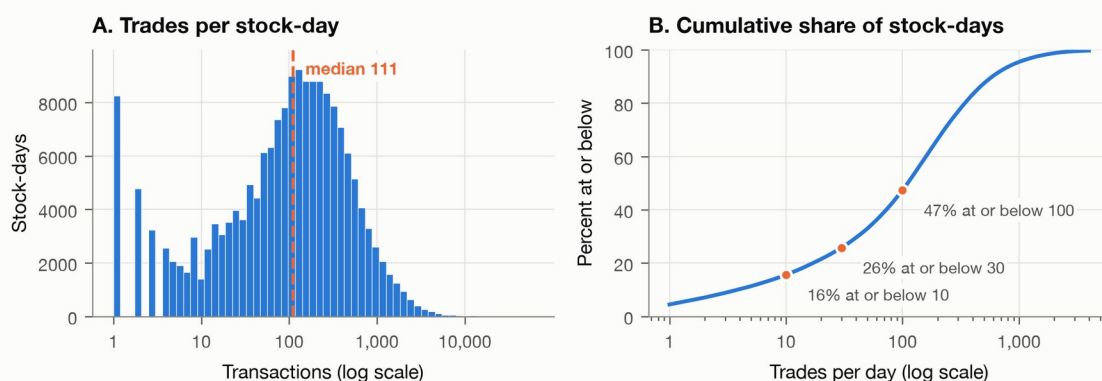


Figure 2. Trading intensity in the pooled exchange universe before instrument filtering. The long low-trade tail helps explain why a naive pooled analysis can look extremely thin, but it should be interpreted together with Figure 1 because the low-liquidity buckets contain many non-equity instruments. Generated by `scripts/03_descriptive.py`.

5.3 Opening auctions, price bands, and observed-price constraints

NEPSE's opening mechanism is economically important. In ordinary equity, the ratio of opening-return variance to close-to-close variance is elevated in the lower-liquidity daily trade-count deciles, reaches its sample maximum in the decile with a median of 79 trades per day, and then generally declines as trading intensity rises. In the thinnest decile (median 24 trades per day), this ratio is 57.4% and 12.7% of opens equal the previous close. Across the full sample, 18.4% of observations in that decile have an absolute opening return between 1.9% and 2.1%. Because the pre-open band widened from approximately $\pm 2\%$ to $\pm 5\%$ during the April 2026 rule transition, this 18.4% is a pooled descriptive fingerprint, not an estimate of the fraction mechanically bound by the former $\pm 2\%$ rule. These patterns are observable features of the auction data, not proof of a latent censored-return model. The project's attempted censored-normal recovery failed its post-April-2026 regime-change check and is therefore not used in this paper.

NEPSE's pre-open call auction is visible in the data

Orders may be placed only within a band of the previous close; if none match, the open is set to that close. Both rules leave clear fingerprints. The band widened from $\pm 2\%$ to $\pm 5\%$ on 2026-04-20; both regimes are shown.

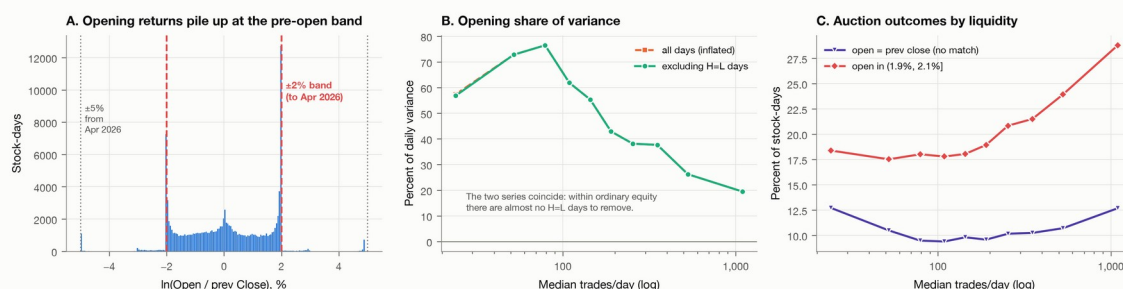


Figure 3. Observable fingerprints of NEPSE's pre-open auction: opening-return pile-ups at the permitted bands, the opening-variance ratio by trading intensity, and the frequencies of no-match openings and near-band openings. The exhibit is generated by `scripts/13_opening_auction.py` and is used descriptively; it is not used to infer a latent uncensored opening distribution.

5.4 The benchmark must match the estimator

A common error is to compare an intraday range estimator directly with close-to-close variance and call the difference 'bias.' Close-to-close returns include the overnight move from yesterday's close to today's open; Parkinson and Rogers-Satchell do not. In the NIFTY 50 data, Rogers-Satchell is only 0.671 of close-to-close variance but 0.965 of open-to-close variance. For the NEPSE thin-equity group, Rogers-Satchell is 0.998 relative to the matched open-to-close benchmark. The decomposition also shows that $\text{Var}(\text{opening return})/\text{Var}(\text{close-to-close return})$ is 63.6% in thin equity versus 26.9% in dense equity. These component ratios are not additive shares because opening and intraday returns can covary; the full identity includes the 2Cov term. Matching the estimator and benchmark to the same time interval is therefore a prerequisite for diagnosing estimator bias.

Rogers-Satchell cannot see the overnight gap, so a close-to-close benchmark conflates two effects

RS is built from open, high, low and close within the session, so comparing it to close-to-close variance mixes estimator error with variation occurring outside the session. Panel B is a ratio, not a share: $\text{Var}(\text{cc}) = \text{Var}(\text{co}) + \text{Var}(\text{oc}) + 2\text{Cov}$.

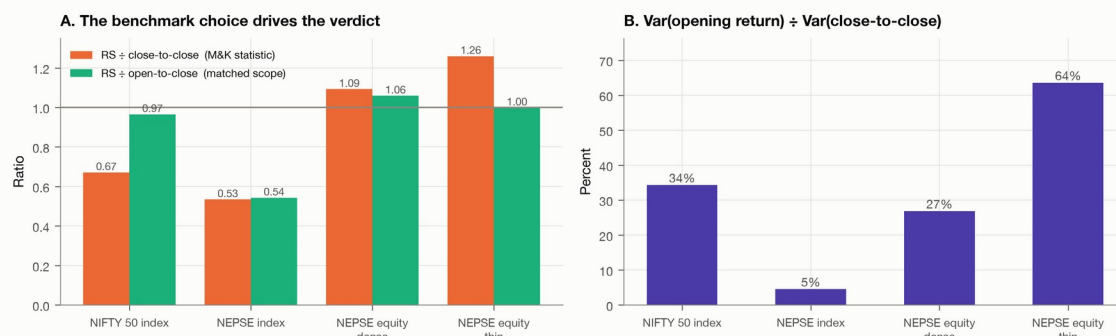


Figure 4. Benchmark scope and variance decomposition. Panel A compares Rogers-Satchell with close-to-close and matched open-to-close variance; Panel B reports $\text{Var}(\text{opening return})/\text{Var}(\text{close-to-close return})$. All ratios are variance-scale and are generated by scripts/12_benchmark_diagnosis.py with numerical output in table18_benchmark_diagnosis.csv.

6. Empirical Results

6.1 Ordinary equities are much more usable than the pooled file suggests

Once the universe is restricted to ordinary equity, the thin-market picture changes sharply. Across the equity sample, median trading intensity is 165 trades per stock-day, the 10th percentile is 37 trades, and only 1.6% of stock-days have fewer than ten trades. The zero-range rate $P(H=L)$ falls from 5.70% in the pooled universe to 0.28% in ordinary equity. The thinnest equity quintile still has a median of 49 trades per day and full participation in the study window.

Table 3. Why instrument classification changes the volatility diagnosis

Measure	Pooled universe	Ordinary equity
Stock-days	184,390	143,149
Median trades/day	111	165
10th percentile trades/day	4	37
Stock-days below 10 trades	14.8%	1.6%
$P(H = L)$	5.70%	0.28%
Parkinson exactly zero	5.70%	0.28%
Rogers-Satchell exactly zero	15.25%	4.35%

The implication is important for a paper about volatility without options: NEPSE does not lack enough price information to measure volatility simply because it is a frontier market. For common equity, the daily range is usually observed with sufficient trading activity to be informative. The data become problematic when different instruments are pooled into the same liquidity ranking.

6.2 Cross-market ratios remain close to one for NEPSE equity

Under identical code, Parkinson's standard-deviation ratio relative to open-to-close variation is 0.978 for the NIFTY 50. Across six NEPSE equity stock-day buckets formed by daily transaction count, it ranges from 0.935 at roughly 33 trades per day to 1.058 around 78 trades per day, and then moves toward 0.972 at the most active end. The relationship is not monotone and, crucially, does not collapse in the least active equity group. The zero-range rate is 1.63% in that thinnest cross-market bucket and zero or nearly zero in the remaining groups.

The same estimator code across three regimes

Range estimators remain usable in NEPSE equity; on NIFTY, the correctly aggregated 21-session Parkinson series still co-moves strongly with India VIX.

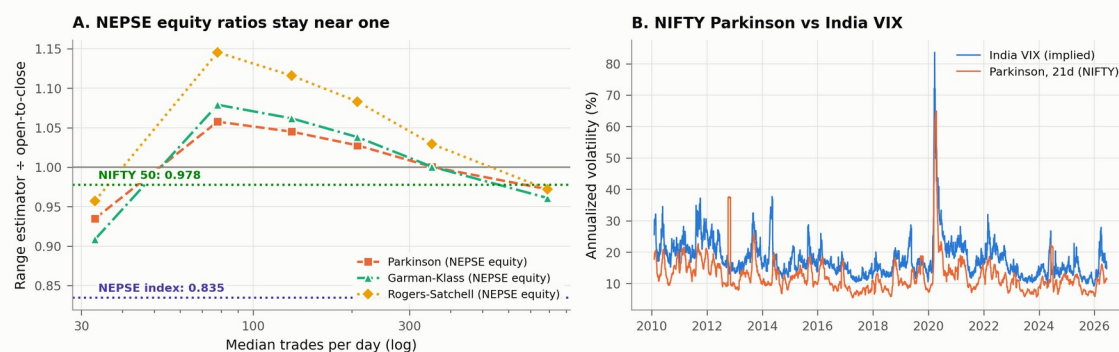


Figure 5. Cross-market estimator ratios under identical code. Panel A shows that ordinary NEPSE equity remains near the matched benchmark across daily trade-count buckets. Panel B compares India VIX with a 21-session Parkinson series constructed by averaging daily Parkinson variance first and taking the square root after aggregation, consistent with Section 7.1.

Table 4. Cross-market fingerprint on a standard-deviation scale

Regime	Median trades/day	Parkinson / OC (SD)	RS / OC (SD)	Zero range
NIFTY 50	Dense	0.978	0.980	0.00%
NEPSE equity - thin	33	0.935	0.957	1.63%
NEPSE equity	78	1.058	1.145	0.00%
NEPSE equity	131	1.045	1.116	0.00%
NEPSE equity	208	1.028	1.083	~0.00%
NEPSE equity	351	1.000	1.029	0.00%
NEPSE equity - dense	789	0.972	0.972	0.00%

6.3 An options-market validation anchor: India VIX

The most relevant external check for the paper's motivating question comes from India. The NIFTY 50 has both cash-market OHLC prices and an options-derived volatility index. To keep the validation

consistent with the paper's reporting formula, the 21-session Parkinson series is computed as the square root of 252 times the rolling mean of daily Parkinson variance; close-to-close volatility is the annualized 21-session sample standard deviation of log returns. On 3,999 matched observations, Parkinson volatility has a correlation of 0.776 with India VIX and an R^2 of 0.602 in a simple linear relationship. The corresponding close-to-close series has a correlation of 0.796 and an R^2 of 0.634. Both measures therefore track the volatility state represented in option prices strongly, but the range-based measure does not outperform close-to-close in this particular validation.

This result should not be interpreted as evidence that Parkinson volatility equals implied volatility. India VIX is forward-looking and risk-neutral; Parkinson is backward-looking and statistical. The useful conclusion is narrower: daily OHLC ranges contain a substantial signal about the same underlying volatility environment that an option market prices. The India comparison supports their use as a practical state measure when no option chain exists, while also showing that range information need not dominate a conventional close-to-close measure in every validation exercise.

6.4 Why an additive correction is not automatically better

The AddRS correction was designed for a setting in which Rogers-Satchell is downward biased because discrete observations miss path extremes. On NIFTY 50, AddRS is 1.005 relative to the matched open-to-close benchmark on a standard-deviation scale, which is reassuring for the implementation. On NEPSE ordinary equities, however, no downward Rogers-Satchell bias is detectable against the available open-to-close proxy: the uncorrected measure is already close to or above that benchmark in every stock-day liquidity quintile formed by daily transaction count. AddRS then rises to 1.149-1.323 of the benchmark. The result is therefore evidence about performance relative to the available proxy, not proof that the estimator's theoretical bias premise is false relative to latent variance.

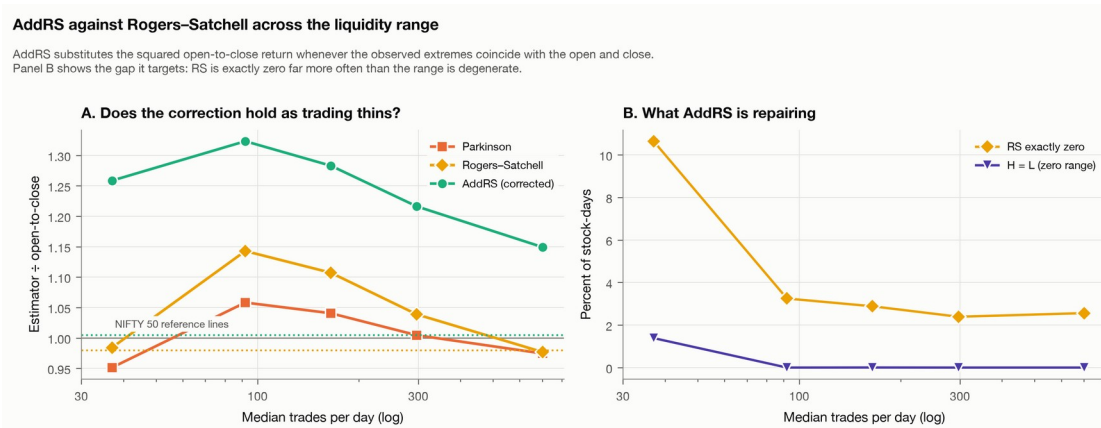


Figure 6. Rogers-Satchell and AddRS relative to a matched open-to-close benchmark. The correction is well behaved on NIFTY 50 but overstates NEPSE equity relative to the available benchmark in this sample.

Table 5. Additive correction relative to the matched open-to-close benchmark (SD scale)

Regime	RS / OC	AddRS / OC	Interpretation
NIFTY 50	0.980	1.005	Correction lands near benchmark
NEPSE Q1	0.984	1.259	No detected RS deficit;

			correction overshoots
NEPSE Q2	1.143	1.323	Base estimator already above benchmark
NEPSE Q3	1.107	1.283	Base estimator already above benchmark
NEPSE Q4	1.039	1.216	Base estimator already above benchmark
NEPSE Q5	0.977	1.149	Correction still overshoots

This is a general lesson for frontier-market work. The AddRS increment is non-negative. Let $x = \ln(C/O)$, and let I_u and I_v denote the upper- and lower-boundary indicators used by the implementation. The code reduces exactly to $\text{AddRS} = \text{RS} + (x^2/2)(I_u + I_v)$, so in expectation $E[\text{AddRS}] = E[\text{RS}] + (1/2)E[x^2(I_u + I_v)]$. Both indicators can fire on a fully monotone bar. Therefore, if Rogers-Satchell is already unbiased for the target variance in a regime and the boundary term has positive expectation, AddRS has positive expected bias relative to that target. Because true latent variance is unobserved here, the empirical conclusion remains narrower: no downward Rogers-Satchell bias is detectable against the matched open-to-close proxy, while AddRS overshoots that proxy in the NEPSE equity sample.

7. A Practical Volatility Protocol for Nepal

The empirical evidence supports a simple but disciplined protocol for researchers, portfolio managers, and risk analysts who need a volatility measure for NEPSE securities without options.

- Define the target first. If the objective is within-session volatility, use an OHLC estimator and compare it to open-to-close variation. If the objective is total daily risk, include overnight variation through Yang-Zhang or a close-to-close component.
- Restrict the universe to the intended asset class. For equity research, classify and retain ordinary equities before ranking securities by liquidity. Do not let debentures, funds, or restricted promoter shares define the 'thin stock' bucket.
- Validate every OHLC bar. Require $L \leq \min(O, C) \leq \max(O, C) \leq H$. Treat negative simplified Garman-Klass or Rogers-Satchell values on invalid bars as data problems, not market phenomena.
- Use the genuine trading-session calendar. Previous close means the previous session, not the previous calendar day. Annualization should use the market's actual session count rather than importing 252 automatically.
- Compute at least two estimators. A practical pair is close-to-close volatility plus a range measure such as Parkinson or Rogers-Satchell. Agreement provides reassurance; disagreement is a diagnostic to investigate.
- Flag $H = L$, zero returns, and low trade counts. These rates should be reported by liquidity group because they reveal where the data cease to describe the latent price path well.
- Do not add a bias correction until the bias premise is supported in the target sample. A correction should be validated against a matched benchmark and, when possible, in a liquid external market.

- For forecasting, separate measurement from prediction. Use the resulting daily or rolling volatility series as an input to HAR, GARCH, CARR-style, or other forecasting models rather than calling the historical measure 'implied volatility.'

7.1 Recommended reporting formula

For a 21-session risk measure, a transparent primary specification for an ordinary NEPSE equity is the annualized rolling Parkinson or Rogers-Satchell volatility:

$$\hat{\sigma}_{21,t} = \sqrt{[A \times (1/21) \times \sum_{j=0}^{20} \hat{v}_{t-j}]}$$

where $\hat{v}$ is the chosen daily variance estimator and A is the stated annual session count or calendar-based annualization factor. The report should show close-to-close volatility beside it and disclose the security's median trade count and zero-range rate over the same window or a longer diagnostic window.

7.2 A suggested hierarchy of estimators

For NEPSE ordinary equity, the evidence in this project suggests the following hierarchy. Use Rogers-Satchell as a primary within-session measure when drift robustness is desirable, with Parkinson as a simple cross-check. Use close-to-close or Yang-Zhang when the overnight component is part of the risk object. Treat AddRS as a model-dependent alternative rather than a default correction. This recommendation is empirical and market-specific; it is not a claim that one estimator dominates in all frontier markets.

8. Discussion

The main result is more optimistic than the usual intuition about frontier markets. The lack of derivatives does not imply that risk cannot be measured, and thin trading does not automatically make daily ranges useless. In the NEPSE ordinary-equity sample, the price path observed through OHLC data contains enough information for range-based volatility to remain near a matched benchmark even at modest trading intensity. This is economically meaningful because it gives a low-data-cost approach to risk measurement in a market where intraday databases and derivative prices are not readily available.

At the same time, the study shows that frontier-market volatility is as much a data-engineering problem as an econometric one. A perfectly coded Parkinson formula can produce a misleading research conclusion if the low-liquidity sample is mostly bonds and restricted shares. A theoretically appropriate Rogers-Satchell comparison can appear strongly biased if it is benchmarked against close-to-close returns that contain a large overnight component. A sophisticated additive correction can worsen measurement if the bias it is meant to repair is not present in the target sample.

The India VIX result is especially useful for interpretation. It shows that the high-low range captures substantial information relevant to an options market's assessment of volatility, which helps explain why range estimators are attractive when options are absent. Under the paper-consistent 21-session aggregation, however, close-to-close volatility tracks India VIX somewhat more closely than Parkinson in this sample, so the external validation does not establish range-estimator dominance. The gap between realized and implied volatility remains fundamental. A future Nepalese derivatives market

would add information that historical prices cannot reconstruct: the market's forward distribution, volatility risk premium, strike-specific skew, and maturity term structure. Until then, a carefully constructed realized-volatility system is a substitute for risk measurement, not a substitute for the entire informational content of options.

9. Reproducibility and Audit Trail

This submission uses only empirical results with a traceable producer in the accompanying paper-specific repository. The reproducibility map links the cross-market fingerprint to script 09, the benchmark decomposition to script 12, the opening-auction diagnostics to script 13, the AddRS comparison to scripts 17 and 19, the universe-composition result to script 22, and duplicate reconciliation to script 24. Frozen CSV tables and figures corresponding to the submitted manuscript are included under output/.

The repository excludes failed and superseded analyses from the larger internal research archive. In particular, this paper does not use the attempted censored-normal estimate of latent opening dispersion, the superseded multi-horizon convergence result, or producerless robustness rows. Results that cannot be regenerated from the retained scripts and audit files are not used as evidence.

For reproducibility, the submission package includes the processed NEPSE stock-day panels, the NIFTY 50, India VIX, and NEPSE index series used by the reported checks, a compact duplicate-key audit extract, a build manifest, and a manuscript-to-output reproducibility map. The original stock-level NEPSE downloads are not publicly redistributed because their redistribution terms remain unresolved; the cleaning/build script and the original aggregate raw-data hash are retained for provenance. All retained empirical claims can be regenerated from the supplied paper-facing inputs.

10. Limitations

Several limitations should keep the conclusions appropriately narrow. First, the NEPSE benchmark is not true integrated variance. Without high-frequency transaction data or an options market, latent daily variance is unobserved. The matched open-to-close squared return is a practical proxy, and Patton (2011) shows that imperfect volatility proxies can affect comparisons. The paper therefore reports whether downward bias is detectable against the available benchmark rather than claiming that the true latent variance is known.

Second, market institutions constrain how prices are formed and recorded. Pre-open bands, daily price limits, circuit-breaker rules, and no-match auction outcomes can change where observed variation appears in the daily bar. The project documents these institutional fingerprints but does not identify the latent opening return that would have occurred without the rules. An attempted censored-normal recovery did not survive the available regime-change check and is excluded from the reported results. Third, the market changed during the sample, including a 2026 trading-week reform and associated rule changes, so future work should model regime shifts more explicitly.

Fourth, the external India VIX exercise is a cross-market validation, not a Nepal-specific implied-volatility comparison. Because consecutive 21-session observations overlap heavily, the reported correlations and R^2 values are descriptive and are not treated as independent observations for inference. A frontier market with different investor composition, price-limit rules, or microstructure could behave differently. Finally, the paper addresses measurement rather than forecasting; out-of-sample performance remains for future work.

11. Conclusion

A frontier market does not need options to calculate a useful measure of volatility, but it does need conceptual discipline about what is being calculated. Without options, Nepal cannot observe option-implied volatility or reproduce the full information in a VIX-style index from spot prices alone. It can, however, estimate historical and realized volatility from daily prices, and OHLC range estimators make better use of those prices than a close-only calculation.

The NEPSE evidence shows that this approach is practical for ordinary equities. Range-based estimates remain close to a matched open-to-close benchmark across liquidity groups, and the thinnest equity bucket does not exhibit the estimator collapse seen in the pooled exchange file. The apparent collapse is largely an instrument-composition artifact. The NIFTY 50 and India VIX comparison further shows that a correctly aggregated rolling Parkinson measure tracks an options-derived volatility state strongly, although close-to-close volatility tracks it somewhat more closely in this sample. This provides external support for the informational value of daily ranges without claiming that they reproduce or dominate implied volatility.

The resulting recommendation is straightforward: classify the securities correctly, clean and validate OHLC data, use the true exchange calendar, match the benchmark to the estimator's time interval, report liquidity diagnostics, and treat bias corrections as hypotheses to validate rather than automatic improvements. For Nepal today, this produces a defensible volatility measure from information the cash market already provides. It does not manufacture implied volatility, but it provides the foundation needed for risk management and for future volatility forecasting in a market without derivatives.

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- Data and reproducibility note. Empirical quantities in this manuscript are drawn from the accompanying paper-specific NEPSE volatility package. Retained claims are backed by producer scripts, frozen tables or figures, or directly auditable source artifacts. The package includes processed paper-facing data, external validation series, a compact duplicate audit, and a reproducibility map; original stock-level source downloads are not publicly redistributed because their redistribution terms remain unresolved. The empirical analyses are descriptive and exploratory unless explicitly stated otherwise.